

Quorum before execution · KAVACHA before damage · x402 inside the agent economy.



A production, live agentic system at the convergence of Agentic AI × DeFi × RWA on Casper.

Casper —
Self-Healing Agentic DeFi Treasury Swarm

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Live: 35.255.196.78/eraya

Demo: 35.255.196.78/eraya_api/demo/

Quorum before execution · KAVACHA before damage · x402 inside the agent economy.

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DeFi does not fail politely.

Humans and single bots break

Treasuries are run by humans or single “trade bots” — both both break under adversarial, real-money conditions.

Failure paths are not optional

On-chain systems punish missing failure paths with irreversible loss, not just a bad log line.

Happy-path demos are not enough

Most agent demos optimise the happy path; none prove what happens when a fix itself fails.

Identity and review are missing

No on-chain agent identity, no economic signal for useful agent work, no adversarial review before high-stakes moves.

ERAYA turns a DeFi treasury into an immune system.



Self-healing treasury swarm

Four specialised AI agents share context, vote on high-stakes moves, and recover rather than failing closed.

Quorum before execution

Specialised agents share context and vote on high-stakes moves before treasury action.

Recovery when fixes fail

If a remediation fails, the swarm re-plans and recovers — a real self-healing loop, not a script.

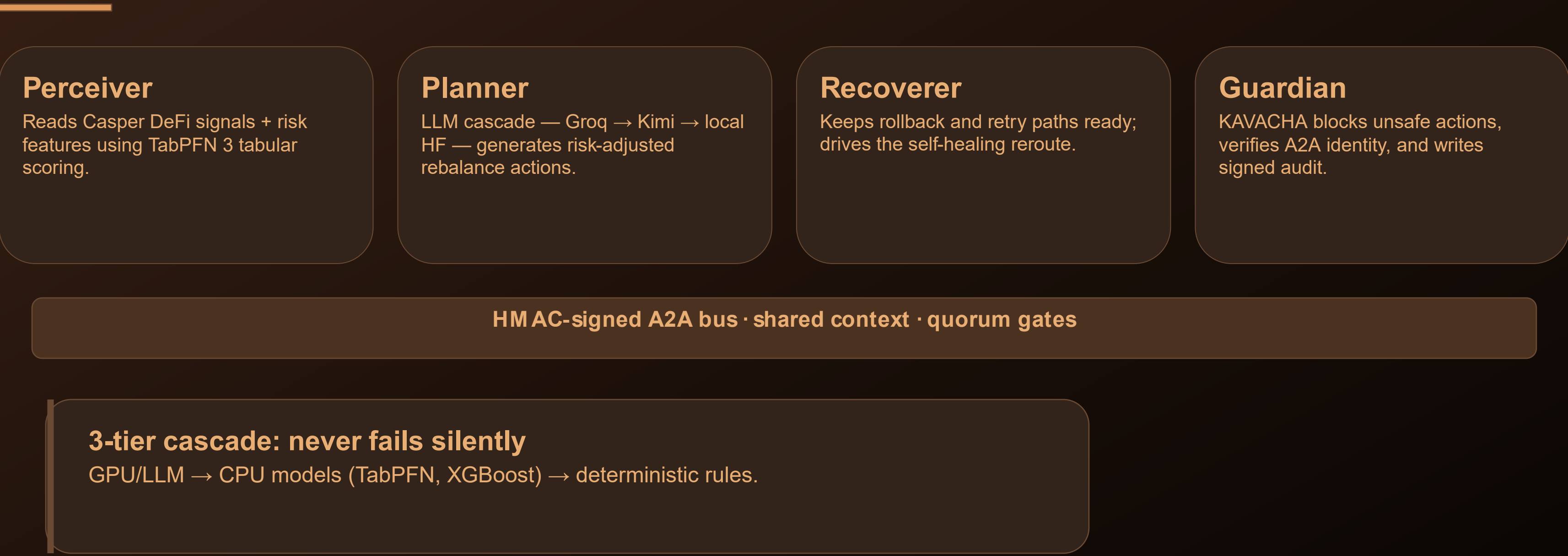
KAVACHA before damage

Security-first by design: KAVACHA blocks unsafe actions and injection before damage.

x402 inside the agent economy

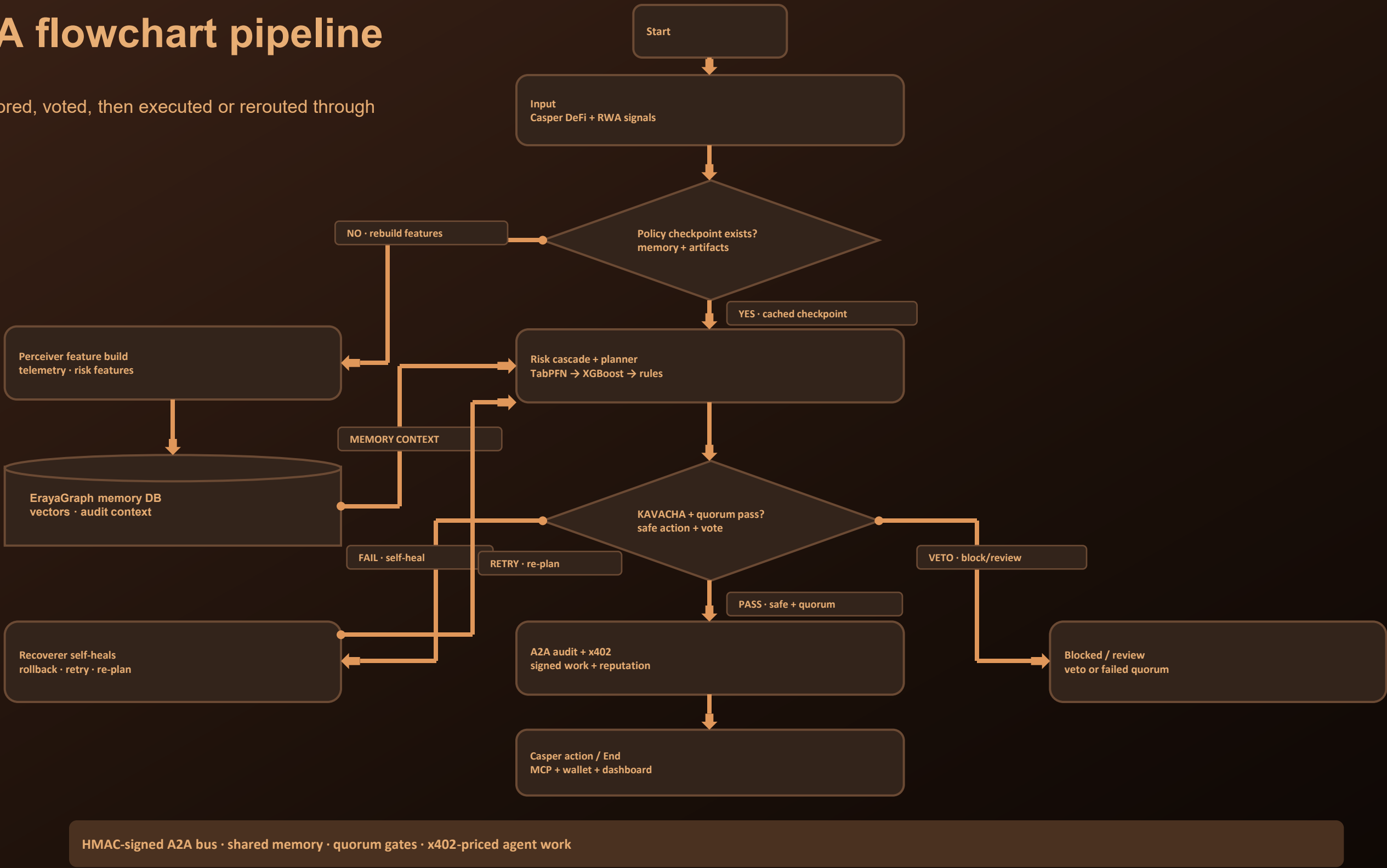
x402 prices inter-agent work; reputation is anchored on Casper.

Perceiver → Planner → Recoverer → Guardian, over an A2A bus.



ERAYA flowchart pipeline

Signals are scored, voted, then executed or rerouted through recovery.



Product, swarm, and Casper layer in one deployed loop.

LAYER	DEPLOYED ROLE
Frontend	Next.js console — DeFi portfolio · threat radar · swarm-consensus UI.
Backend	Django + Channels — REST/WebSocket, audit log, domain registry (daphne on the VM).
Swarm core	Perceiver · Planner · Recoverer · Guardian, coordinated via HMAC-signed A2A + ErayaGraph shared memory (Pinecone-backed).
Casper layer	MCP facades · CSPR.click wallet · x402 + reputation.
Observability	OpenTelemetry traces on every cascade tier.

Casper-native, now writing to the chain.

casper_defi domain adapter

Live CSPR.cloud / CSPR.trade-shaped telemetry; the live validator set feeds decentralization detection.

pool TVL APY gas slippage liquidity depth governance quorum

validator uptime mempool

x402 payments (LIVE)

HTTP-402 challenge → pay
→ 200 flow served for paid
market data.

CSPR.click + MCP

CSPR.click wallet + Casper
MCP facades — graceful
demo→live switch.

Odra contracts (next)

Agent registry, treasury,
governance, and reputation
on Casper testnet.

On-chain anchoring

Critical anomalies anchored
as real Casper 2.0 testnet
transactions via casper-client
put-transaction — evidence
hash verifiable on
testnet.cspr.live.

A priced, audited, voting swarm — not a single DEX-calling agent.

LLM cascade

- 1 Groq (llama-3.3-70b)
- 2 Kimi for long-context planning
- 3 Local Hugging Face fallback

Every provider is a graceful-fallback facade — the demo never breaks if a key/service is down.

Risk brain

TabPFN 3 tabular foundation model (live, hosted) scores DeFi risk with zero training.

Memory / RAG

Pinecone vector store (ErayaGraph); Firecrawl web ingestion.

Multilingual + orchestration

Sarvam AI for multilingual support; LangGraph swarm graph for orchestration.

Economic signal

x402 prices inter-agent work while audit and quorum preserve trust.

Threats hunted before a bad deploy lands — and anchored on-chain.

KAVACHA

Security-first guardian: blocks unsafe actions and injection, verifies agent identity, and anchors critical findings on-chain.

Critic agent (5th reviewer)

Independent LLM-as-judge scores every high-stakes action
APPROVE / REVISE / REJECT before execution.

Injection kill-shot

Unsafe inputs BLOCKED with a signed audit trail — live: rule R003, full timeline.

A2A identity defense

Forged agent messages rejected via HMAC — same function as the live path.

Quorum before execution

High-stakes moves need swarm consensus, not one agent's call.

Network Anomaly Intelligence

Statistical detection over mempool, governance & validator telemetry — z-score baselines, MEV correlation, Sybil clustering (Gini/entropy), Nakamoto coefficient; findings anchored on Casper testnet.

Serving live — and writing real transactions to Casper testnet right now.

http://35.255.196.78/eraya_api/demo/

\$820K TREASURY	12.86% APY	47% GOVERNANCE QUORUM	99.2% VALIDATOR UPTIME	TabPFN RISK SCORE	200 DASHBOARD API
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Real on-chain proof

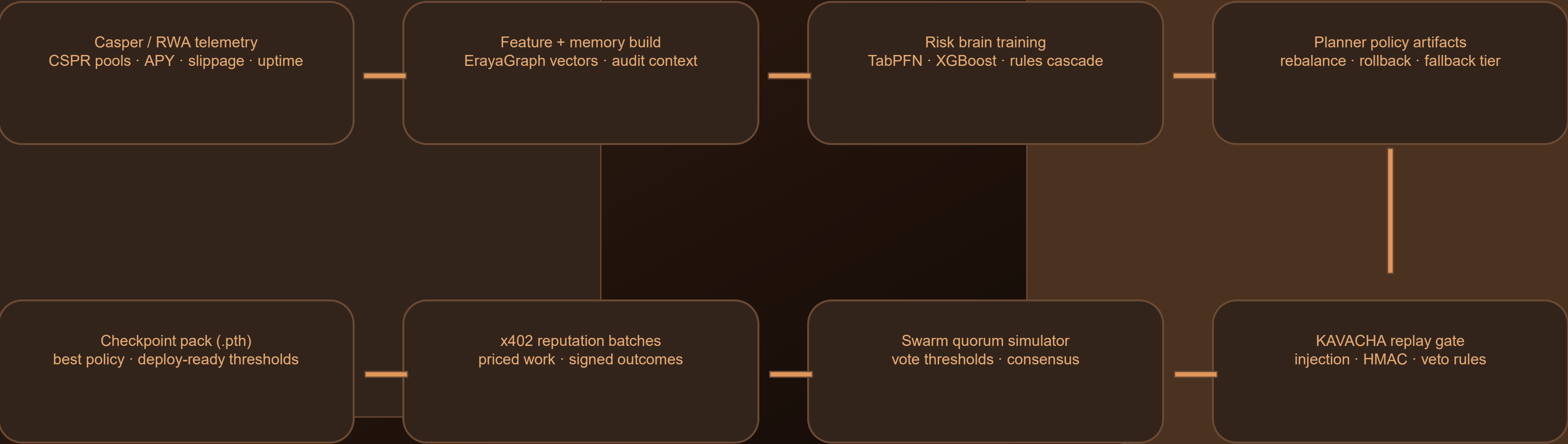
"Run Anomaly Scan" detects a governance_sybil / validator_centralization anomaly and anchors it as a real testnet transaction — verifiable on testnet.cspr.live.

Production, not slideware

Real provider keys (Groq, Kimi, Pinecone, Firecrawl, TabPFN, Sarvam) + a funded Casper account.

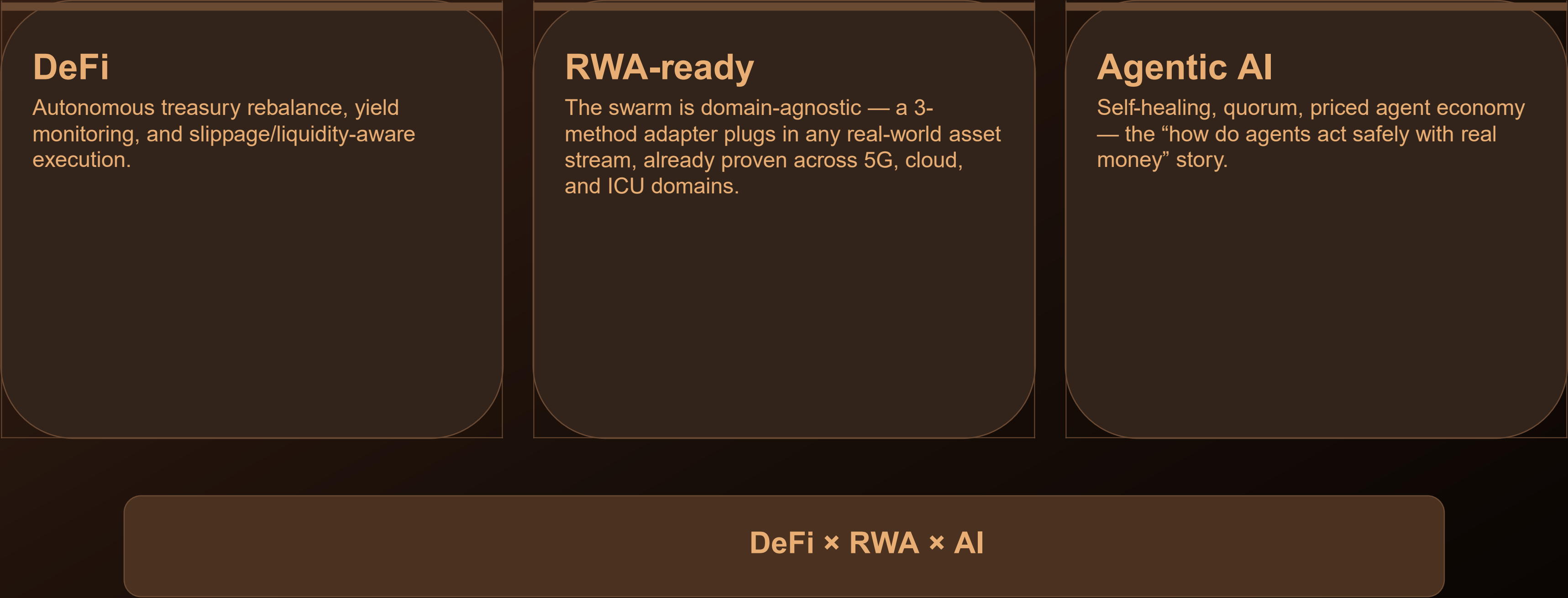
ERAYA model artifact pipeline

Turns Casper telemetry, DeFi/RWA events, and safety replays into deployable swarm policies and audited checkpoints.



Artifact loop: live providers → cached features → safety replay → quorum simulation → deployable ERAYA policy pack

One engine, three of Casper’s priorities.



Defined failure behaviour is our moat.

Every agent has a defined failure path

The swarm never fully stops — it can re-plan, retry, roll back, or fail safely.

Actually on-chain

Anomalies are anchored as real, verifiable Casper testnet transactions — not "coming soon."

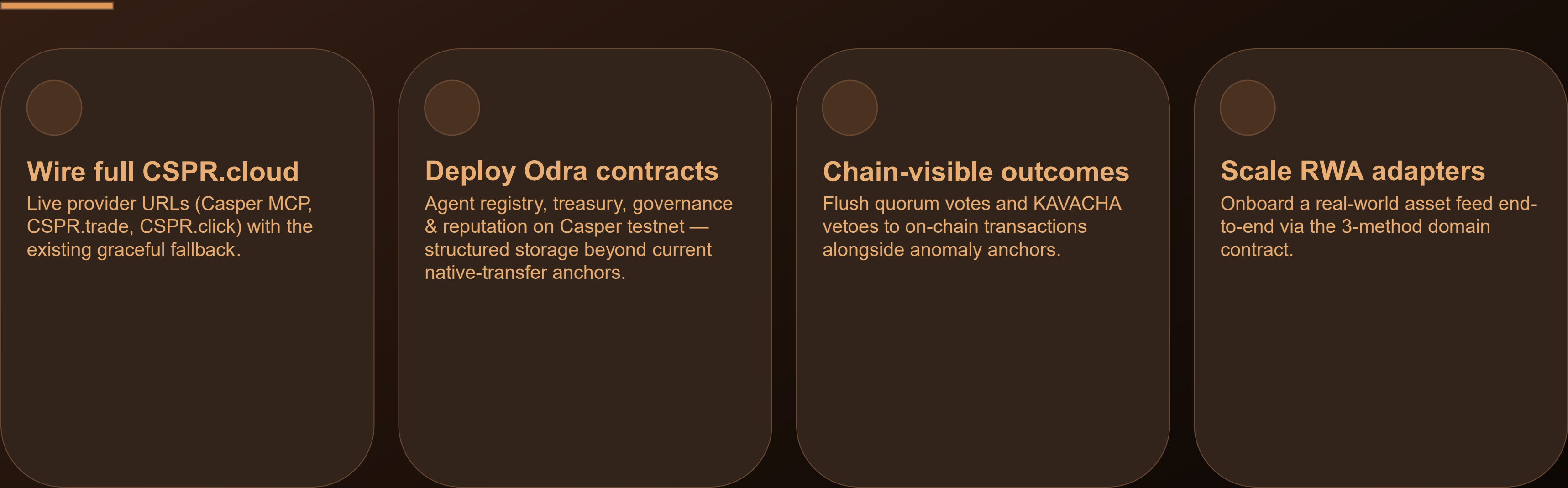
Real signals on the critical path

Real LLM planning + real TabPFN risk + real on-chain writes + real x402 payments — no stubs on the critical path.

Two hard problems solved together

Agent safety (KAVACHA + statistical anomaly intel + Critic) + agent economics (x402 + on-chain anchoring).

Anchoring, x402, TabPFN & Critic are done — next, deeper on-chain execution.



Done: anchoring · x402 · TabPFN · Critic → Next: Odra · full CSPR.cloud · chain-visible outcomes · RWA scale

Team ERAYA

Team Lead

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Interactive demo

http://35.255.196.78/eraya_api/demo/
Console: 35.255.196.78/eraya · On-chain proof: anomaly-anchor transactions on testnet.cspr.live.

Repository

github.com/martian3062/eraya_microsoft
branch caspr · open-source

The ask

Vote ERAYA Casper on CSPR.fans to advance to the Final Round — a live, self-healing agentic DeFi treasury that actually writes to Casper.